Luke Miller

12/5/2022

Health Research Paper

Abstract:

Research has shown that propensity for technology adaptation varies greatly between states, and that states that adopt technology quickly tend to do so consistently across fields (medicine & agriculture), and over time (patterns maintained over 50 years). This paper seeks to explore the role that state institutions play in technology adaptation by observing 2 public health laws relating to emerging technologies & emerging societal problems. I use a simple cross-sectional regression to explain the lags in Good-Samaritan Laws & in Laws Expanding Access to overdose prevention medication (Naloxone) in the context of the opioid epidemic. I find that the length of delay in the passage of legislation expanding access to life-saving medication is associated with a higher incidence in Overdose Rates at the time legislation is passed. And earlier passage of legislation is positively associated with earlier passage of similarly targeted legislation.

And, I find that longer delays in access legislation are correlated with a more right-leaning voter base & higher state median incomes. Lastly, I regress previous technological adaption on the and find these associations largely hold with technologies less directly correlated with state government action. I conclude that a state's propensity to adopt new technologies does depend on state-level characteristics rather than solely the potential benefits of the technology.

Importance:

This paper is inspired by the work of Jonathan Skinner and Douglas Staiger who asked why certain states and institutions are more likely to adopt highly cost-efficient and effective medical technologies (Skinner & Staiger, 2003; Skinner & Staiger, 2015). Their findings offer explanations as to why medical care & costs vary widely by state, even for differing treatments. But they also address the divergent literature on overarching causes for varying diffusion rates. Differing views on the causes of variation of diffusion rates of technology have persisted in a debate dating back at least to Zvi Griliches' 1957 study on technological diffusion in agriculture. He found that Hybrid Corn is most likely to be adopted where land is less fertile, and thus, where the marginal returns to adoption are high (Griliches, 1957). Griliches argues that profitability & higher marginal benefits are the impetus for technology dissemination. This view is contrasted by Everett Rogers' theoretical framework in "Diffusion of Innovations" which outlined varying sociological factors ranging from perceived advantages to social awareness, product lobbying, and state-level decision makers (Rogers, 1962). My paper investigates the role of the latter, state-level decision makers. I focus on the State's decision to adopt technology through legislative action as a way of understanding one mechanism for why highly effective technologies may disseminate slowly.

The primary technology I investigate is Naloxone, an opioid receptor antagonist. This medication rapidly reverses the effects of opioid drugs, quickly restoring normal respiration. Naloxone is intriguing in that it has been available since the 1960's but was slow to be deployed in the opioid epidemic, even for first responders (Davis et al.,2014; Davis et al.,2013). This is a unique technology in that it is the only medication that can effectively reverse an overdose from opioids, in a time where demand for a solution to the epidemic was - and still is - burgeoning (Tracy et al., 2005; Davidson, 2003). Opioid-related deaths in the United States have exploded from <10,000 in 1999 to ~68,000 in 2020 (CDC Wonder). Thus, it is a prime example of a highly effective technology that has been slow to disseminate.

> *I, Surgeon General of the United States Public Health Service, VADM Jerome Adams, am emphasizing the importance of the overdose-reversing drug naloxone. For patients currently taking high doses of opioids as prescribed for pain, individuals misusing prescription opioids, individuals using illicit opioids such as heroin or fentanyl, health care practitioners, family and friends of people who have an opioid use disorder, and community members who come into contact with people at risk for opioid overdose,* ***knowing how to use naloxone and keeping it within reach can save a life.***
>
> **BE PREPARED. GET NALOXONE. SAVE A LIFE.**

HHS.gov

While Naloxone is not a new medication, new improvements have been made such as Evzio's production of an auto-injector (2014) & Narcan's production of a nasal spray (2015) – making it accessible to the lay-person. Furthermore, the risks associated with Naloxone are very low, with the only contraindication (reason to not administer) being an allergy to the medication itself. Which implies that it is safe to use in the event that an individual is falsely assumed to be overdosing on Opioids. Thus, with such high potential and little chance of causing direct harm, questions as to why this product may not be deployed rapidly may be answered by societal pressures & state regulations.

One potential rationale for slow dissemination may arise from a paternalistic approach implying that any action supportive of drug-users may encourage further drug use. Currently, there is heated debate in the literature as to whether a harm-reduction approach to opioid use does create a moral hazard, enabling further use. Recent controversial papers by Packham and by Doleac & Mukerjee (not yet published) have found substantial increases to risky behaviour (up to 21% increase in heroin use) resulting from safe injection sites, needle exchanges, or naloxone access (Packham, 2022; Doleac & Mukherjee, 2021). But this has been a highly contentious area for practitioners, and contradicting peer-reviewed results show that access to Naloxone is unassociated with increases in risky behavior, and that training with naloxone decreases use (D Jones, 2017; Wagner, 2010).

Beyond the moral hazard debate, the use of a cost-benefit analysis would be poorly applied given the substantial impact in reducing overdoses relative greatly outweighs the unsubstantiated increases in use. Expansion and other methods to increase access have been shown to be highly effective in its goal of reducing deaths from Opioids (Chimbar & Moleta 2018). Models in the UK have suggested that active distribution of Naloxone would decrease Overdose deaths by around 6.6% per year (Langham, 2018).

Thus, this paper moves beyond exploring the effectiveness of the treatment, instead, asking why the treatment didn't expand more rapidly. Thus, I explore the impact of the current in-state impact of the opioid epidemic, political leanings, passage of previous legislation pertaining to the epidemic, and the professionalism of state legislatures on the delayed legislation expanding access to Naloxone for “take-home” use. The 2 specific legislation types we will examine are Naloxone Access Laws and Good-Samaritan Laws. Naloxone is not approved for over-the-counter use by the FDA, but as of 2021 49 of 50 states have passed legislation to allow individuals to buy Naloxone without a prescription (PDAPS; Kimball, CNBC, 2022). Meanwhile, Good-Samaritan Laws are implemented to encourage the use of Emergency Medical Services by providing immunity for opioid possession or usage and predates the Naloxone access laws for all states. Thus, we use it as a benchmark for the state’s previous interests in addressing the opioid epidemic.

This paper is also inspired by the findings of McLellan, who compared the 30 states that had passed legislation in 2017 to those who had not yet and found that early implementers of Narcan Access & Good-Samaritan Laws were associated with much lower opioid-related overdose rates (McLellan et. al, 2016). However, by Griliches’ theory on the dissemination of technology, we would anticipate naloxone use to be high in states where the need is greatest. The contradiction with the legislature provides potential intuition as to the role of state actors in the diffusion of technology. This paper hopes to expand on the findings above, contributing to the literature on technological diffusion by exploring why affordable technologies ($5-$120 / dose) that are highly effective and urgently needed, may not be deployed to the areas where the need is highest and the marginal returns greatest. Specifically, how the state government’s actions and political motivation shape technological diffusion.

<u>Assumptions</u>:

While not integral to the analysis, I assume that there are specific technologies that are reliant on state-level approval for “authorization” to disseminate. This paper assumes that the state legislature’s action to expand access to Naloxone - a life-saving opioid antagonist – is determinant in how quickly Naloxone is distributed to those in need. While not essential to the analysis of the data, this assumption lies at the core of the paper, assuming that the legislature’s actions directly impact the dissemination of the technology (Naloxone). And, that the dissemination of certain additional technologies would similarly be subject to the same legislators.

If, as Griliches did, we suppose that the state acts at the aggregate to either maximize agriculture via hybrid corn adaptation or to decrease overdose deaths by increasing access to Naloxone, then we would expect Naloxone should be quickest to disseminate in the states where the Opioid Epidemic is worst. And conversely, slowest where the Opioid Epidemic is less affected. With my previous assumption that state interests would align with the individuals, we would expect legislation expanding Naloxone access to be correlated with higher rates of overdose. My analysis is largely consistent with the findings of Mclellan, where lag in legislation is positively correlated with overdose rates. But, as discussed further in the limitations section, when looking at only the fastest and slowest adopters, we do see early adoption of the most heavily impacted. But, beyond the first wave of innovators, the association reverses. This describes a situation that the earliest to adopt are those who are, in fact, desperate, but that other societal pressure may impact further rollout.

<u>Data & Cleaning:</u>

I focus mainly on 2 dependent variables; legislation lags (in days) for 2 separate state-level laws. The lag is measured relative to the first state's passage. Information for both laws are available through the Prescription Drug Abuse Policy System (PDAPS). I use this information to create a new data set whereby the first state to adopt the legislation is set to day zero and all legislation's lags are given relative to the first. Because of the variation in language and the amendments to each law, I determine Naloxone Access to have been passed when the state makes it legal for Naloxone to be obtained without a prescription. Abouk, (2019) found that by narrowing the definition of legislation further to the manner with which access was administered (by granting pharmacist powers to distribute) there exists a significant decrease in fatal overdoses. That is, the legislation was much more effective in those states.

But for this analysis we are interested in efforts by the government to expand access, rather than the impacts of the laws themselves.

Similarly, Immunity Laws vary widely in the type of immunity offered and to whom it is offered when a call for EMS is made. Therefore, I generalize Immunity laws to encompass ANY Good-Samaritan Legislation addressing the presence of Opioids. The implementation of these laws span from 2007-2021. There is 1 state without an Access law & 3 states without an immunity law. Rather than exclude these as they will contain the most extreme values if/when legislation passes, I set the passage of their values = to todays date (12/7/2022).

A secondary analysis will examine if previous technologies may be explained by the same independent variables to see if a similar pattern emerges. Previous technologies include the Year that Hybrid Corn Use achieved 10% in the 1950s, the % of Households with a computer in 1993, and the % of states using Betablockers to respond to Acute Myocardial Infarctions in 2000. This data is all estimated from the figures in Skinner & Staiger (2003) due to time constraints for this course.

The analysis includes a variety of independent variables but emphasizes the state of the opioid epidemic to explain demand for medication, proxied by the overdose rates at the time legislation was passed. Age-Adjusted Overdose rates from 2013-2020 are sourced by CDC Wonder Data. Further explanatory variables to address state & legislative culture include the Voter Political Leaning Index (2018) & the Squire Professionalism Index (Squire, 2017). While neither measure gives “interpretable measures” they do offer a sense of the culture and attitudes of the state and it’s decision-makers. The Political Leaning Index is created by Cook Political’s polling of voter self-perspectives. It’s scale ranges from (-50 to +50) where more negative is associated with more liberal attitudes and more positive is associated with more conservative beliefs. The Squire Index was created by Peverill Squire in 2015 to address professionalism of state legislatures by evaluating staff size, salaries, days spent in session, and an overall ability to facilitate legislative action. His measure is scaled from 0 to 1, where 0 represents the least professional government and 1 the most effective. Finally, I include measures of household median income available publicly through the Census to correct for associations between incomes and political leanings.

<u>Methods:</u>

For the first iteration of this paper, I will be utilizing a simple cross-sectional approach (1) to identify associations that may explain the speed that a state implements legislation. Each observation is said to occur the year legislation in each state is enacted. Thus, the observations will span across t. Given this is a cross-sectional model I take additional precautions to observe how the trends or early impacts of the opioid epidemic may have influenced the legislation as well by considering the rates in 1999 and in 2005 (2 & 3).

(1) $Y_i = \alpha + B*OverdoseRate(whenpassed)_i + Theta*X_i + U_i$

(2) $Log(Y_i) = \alpha + B*OverdoseRate1999_i + Theta*X_i + U_i$

(3) $Log(Y_i) = \alpha + B*OverdoseRate2005_i + Theta*X_i + U_i$

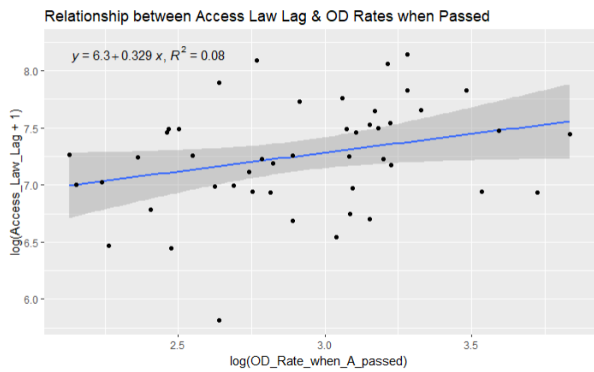
I then utilize the primary model (1) to explore whether past technology adaption can be explained by the same explanatory variables. This is done under the assumption that relative median incomes (from 2017), political leanings (from 2018), and state legislature professionalism (from 2015) have remained relatively constant from 1993 to 2020. While it is plausible to assume a generally consistent state landscape in politics, incomes, & legislature from 2014 to 2020, this is a bold and likely incorrect assumption when applied to larger time frames. <u>Ultimately, the assumption is made for the timely inclusion of these estimates in the paper for this semester's course</u>. I also examine the relationship between these independent variables and hybrid corn adoption in the 1940's & 1950's. Unsurprisingly, results show these relationships weaken as the time differential grows.

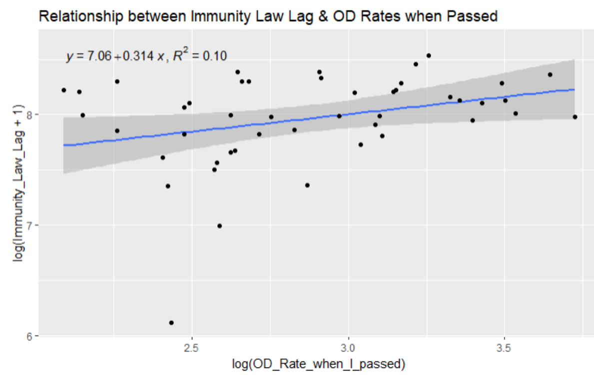
Future iterations will implement a duration model, which better describes the amount of time that elapses until a given event, but will require a significant re-evaluation of the way variables are constructed as the lagged days in legislation currently approaches a continuous measure while many of the measures are available only as a fixed effect or are only observable annually or bi-annually. By using a duration and hazard model, we may be able to observe the speed that one mitigates damage as a function of the state overdose rate. Ultimately, this may offer more insight and greater strength behind many of the conclusions seen above.

I then offer some limitations and additional considerations that offer deeper insight into my findings in addition to pushing future research in a new direction, and which support the above suggestion for evaluating within a duration model.

Results:

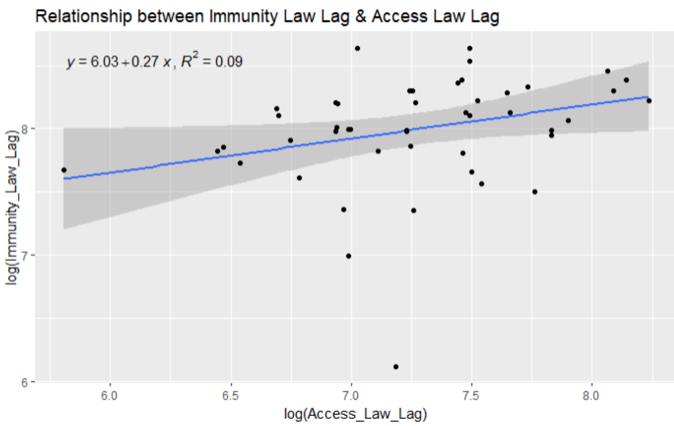
First, we investigate the findings of McLellan. McLellan found that early implementers had 15% (p=.03) lower overdose rates for the earlier implementers of Immunity legislation. And 14% (p=.05) lower overdose rates for early adopters of Naloxone Access Laws (Mclellan et al., 2016). But their study, a 2 sample comparative analysis in 2017, does not account for endogeneity from the impact of the legislation on the earliest implementers. We find very similar results when looking at the rates of overdose at the time of passage, with one important caveat. States that implement overdose mitigating legislation later were likely to have higher overdose rates at the time of passage (see figures below). What we do not see is a persistent negative correlation between overdose rates and lag in passage. This could suggest that government action may not always function in agreement with the technological dissemination theory of Griliches. And, this relationship is maintained when accounting for state-level controls. However, when we break our observations into quintiles and analyze separately, the earliest implementers of this legislation had higher relative rates at the time they passed expansion legislation. This potentially contradictory finding is discussed further in the limitations section, but in short, it is due to using cross-sectional analysis at different points in time while overdose rates are trending upwards.





	Access Law Lag			Access Law Lag			Access Law Lag			Access Law Lag			Access Law Lag		
Predictors	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*
(Intercept)	517.76	891.40	0.564	836.57	890.24	0.352	1045.92	763.02	0.177	415.17	844.66	0.626	656.35	469.87	0.169
OD Rate when A passed	23.57	11.91	0.054	22.33	12.14	0.073	22.50	11.91	0.065	23.76	11.78	**0.050**	23.46	11.76	0.052
Immunity Law Lag	0.15	0.09	0.095				0.17	0.08	**0.047**	0.15	0.08	0.090	0.15	0.08	0.085
political lean	7.98	7.03	0.262	10.95	6.95	0.123				9.26	6.18	0.141	7.38	6.16	0.237
professional	-446.38	1127.84	0.694	-497.71	1151.60	0.668	-1036.78	1004.22	0.308				-438.31	1114.54	0.696
MedianIncome	2.30	12.53	0.855	4.90	12.70	0.702	-4.30	11.14	0.701	2.11	12.40	0.866			
Observations	49			49			49			49			49		
R^2 / R^2 adjusted	0.206 / 0.114			0.152 / 0.075			0.182 / 0.108			0.203 / 0.131			0.206 / 0.133		

After including controls for lean, income, professionalism, and previous legislation, we see that the state OD Rate in the year of passage maintains its large positive impact on the delay in legislation. As stated previously, the political lean and professionalism measures included in the OLS regression are not easily interpretable numerically. But the overall signage of the variable is consistent and provides for useful insight. The impact of a more professional government is likely to increase speed legislation is processed and passed. And more conservative states could be more apprehensive to adopting naloxone without understanding the implications on generating further drug use, reminiscent of the moral hazard debate. After the inclusion of several controls, the regression above indicates that a 1 death per thousand increase in the overdose rate is associated with between a 22.3 – 23.8 day delay in the passage of naloxone access legislation.



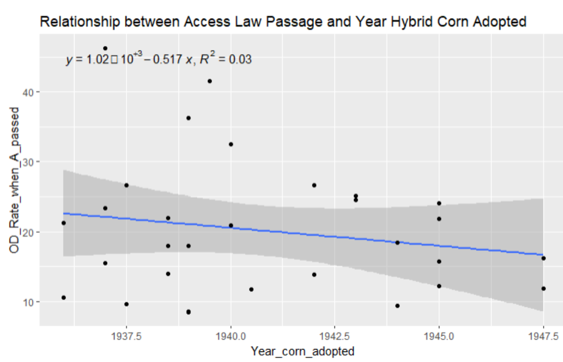
A separate and logical finding is that early legislation pertaining to opioids may encourage further legislation on a similar topic. In some way this may speak to a "momentum" based approach to understanding legislation. While the findings are not statistically significant when controlling for political lean, professionalism, and Median Income, the figure below does show that a 10 day lag in the passage of Immunity legislation explains a 1.5 day lag in the Access Law Passage. It is consistent (between .14 to .17) despite not being significant with any combination of controls present.

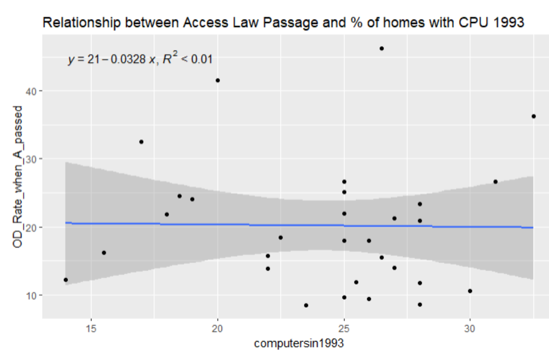
For the next phase of analysis, using models 2 & 3, I regress the lagged legislation onto the state's earliest rates of overdoses to understand if mitigating legislation is better viewed as momentum based – and a function of historical exposure. However, I find no correlation between rates in 1999 and 2005 with the lagged passage of overdose legislation from 2017-2022.

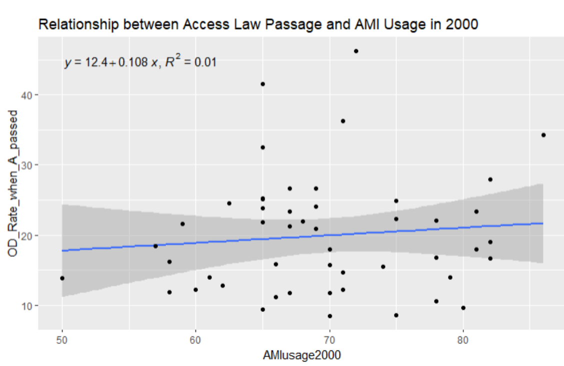
	log(Access Law Lag+1)			**log(Access Law Lag+1)**			**log(Access Law Lag+1)**			**log(Access Law Lag+1)**		
Predictors	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*
(Intercept)	7.34	0.76	**<0.001**	7.23	0.57	**<0.001**	7.84	0.72	**<0.001**	6.90	0.63	**<0.001**
OD Rate 1999	0.03	0.06	0.569	0.04	0.06	0.526	0.00	0.06	0.936	0.03	0.06	0.602
Immunity Law Lag	-0.00	0.00	0.831				0.00	0.00	0.957	-0.00	0.00	0.837
political lean	0.02	0.01	0.091	0.02	0.01	0.090				0.02	0.01	**0.013**
professional	-1.86	1.78	0.302	-1.86	1.76	0.297	-3.39	1.58	**0.037**			
Observations	50			50			50			50		
R^2 / R^2 adjusted	0.153 / 0.078			0.153 / 0.097			0.097 / 0.039			0.133 / 0.076		

	log(Access Law Lag+1)			log(Access Law Lag+1)			log(Access Law Lag+1)			log(Access Law Lag+1)		
Predictors	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*
(Intercept)	7.55	0.82	**<0.001**	7.39	0.65	**<0.001**	7.92	0.80	**<0.001**	7.06	0.66	**<0.001**
OD Rate 2005	0.00	0.04	0.950	0.00	0.04	0.909	-0.00	0.04	0.930	0.01	0.04	0.880
Immunity Law Lag	-0.00	0.00	0.746				0.00	0.00	0.986	-0.00	0.00	0.768
political lean	0.02	0.01	0.112	0.02	0.01	0.116				0.02	0.01	**0.014**
professional	-1.80	1.79	0.320	-1.79	1.77	0.319	-3.37	1.54	**0.034**			
Observations	50			50			50			50		
R^2 / R^2 adjusted	0.147 / 0.071			0.145 / 0.090			0.097 / 0.039			0.128 / 0.071		

For the final bit of analysis, and the ultimate goal of this investigation, we explore whether there exists a relationship between legislation adoption rates and adoption of a variety of previous technologies ranging from the 1950's, 1993, and 2000; which would speak to a consistent relationship between government action and technologies in general. However, there is ultimately no verifiable statistical relationship between Naloxone legislation and past technologies.







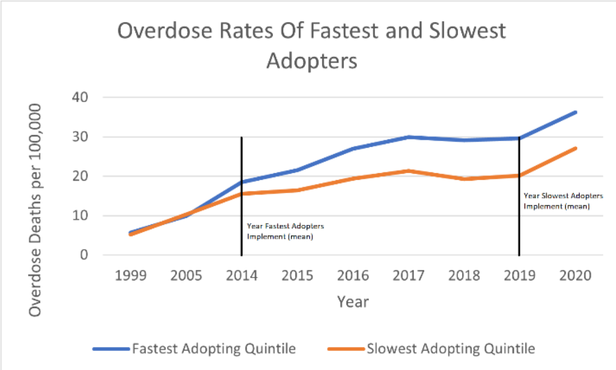
Instead, we examine whether some of the factors that determined legislation passage rates could explain previous technological adoption. I find that factors that had a significant relationship with the speed legislation was passed appear to have a similar relationship with past technology adoption (See Regression Tables Below). Notably, that right leaning states tend to be slower to adopt technology. Ultimately however, accounting for incomes, we see that the interaction between political leaning and median income could easily be overlooked. And simultaneity issues are bound to occur; where income may impact political leaning and political leaning may impact income, while both may influence propensity to use newer technologies. Thus, the take-away may be that rich states have higher adoption rates - a recursive problem in itself that may better be explained by an economic model detailing divergent growth.

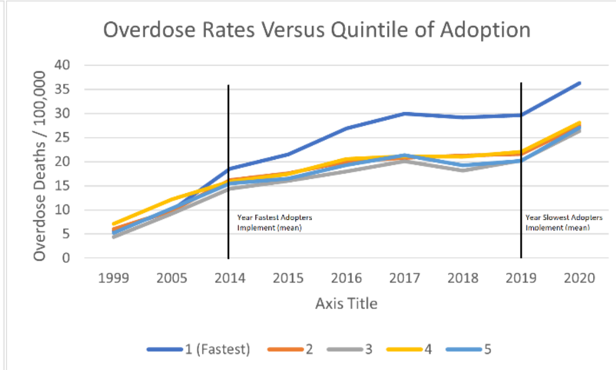
	AM Iusage 2000			**computersin 1993**			**Year corn adopted**		
Predictors	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*
(Intercept)	50.93	21.72	**0.023**	-26.10	13.88	0.071	1947.89	16.40	**<0.001**
political lean	-0.04	0.06	0.477	-0.02	0.05	0.700	-0.01	0.06	0.809
Immunity Law Lag [log]	-0.41	2.34	0.863	3.32	1.95	0.100	0.75	2.31	0.746
MedianIncome	0.36	0.13	**0.010**	0.41	0.08	**<0.001**	-0.22	0.10	**0.031**
Observations	49			31			31		
R^2 / R^2 adjusted	0.258 / 0.209			0.711 / 0.678			0.276 / 0.196		

	AM Iusage 2000			**computersin 1993**			**Year corn adopted**		
Predictors	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*	*Estimates*	*std. Error*	*p*
(Intercept)	81.82	19.66	**<0.001**	-35.96	18.59	0.063	1953.28	17.41	**<0.001**
political lean	-0.14	0.06	**0.017**	-0.20	0.04	**<0.001**	0.09	0.04	**0.046**
Immunity Law Lag [log]	-1.40	2.46	0.572	7.78	2.34	**0.003**	-1.68	2.20	0.449
Observations	49			31			31		
R^2 / R^2 adjusted	0.138 / 0.101			0.450 / 0.410			0.137 / 0.075		

Limitations:

A major consideration for future and expounding research, would be to consider relative overdose rates compared to nominal. That is, to view the overdose rates at time of passage relative to the national average, accounting for the trended growth of opioid deaths. It is possible that the lates adopters had the highest rates compared to when the first states adopted. But that when they adopted, they were actually relatively lower.





In fact, the earliest implementers (quintile 1) had the highest **relative** rates, especially visible when compared to the 5th quintile. But, the trend dissipates for diffusion among quintiles 2 to 5. Ultimately suggesting that there is some support for Griliches' theory – that the first states to implement technology are those with the greatest need - such that the **origin** of technological dissemination follows highest need. But, the diffusion across the remaining 40 states are subject to sociological factors that we do not yet fully understand.

There is clearly more insight that can be gained by examining the same regression where overdose rates at time of passage are instead measured as relative to the national average – approaching a Difference in Differences design. However, these figures above do not dispute the notion that late implementers waited for their situation to be much worse than the first implementers prior to acting. On average, the legislation in the 5th quintile wasn't passed until they reached levels of the first quintile.

<u>Conclusions:</u>

As stated above, this paper does not make strong conclusions about the characteristics of the states that adopt technology quickly. To do so would require a more advantageous model - that is hopefully forthcoming. What this paper does do is offer further insight to the debate between Griliches' and Rogers' Theories of Diffusion. Even should diffusion occur most rapidly in the states of highest need, further sociological factors may prevent need from being the primary impetus for passage of legislation beyond the initial implementers.

There are many factors that determine the diffusion of technology, including state-level actors. These actors are influenced by the state-level incomes and budgets, political motivation, legislative culture, and desire to address arising problems. To some extent, we can see persistent patterns in the states that adopt technologies based on their cultures or beliefs. But, in future research, I would like to investigate the interaction between incomes and technology adaption, ceteris paribus. And to the extent that the state's actions are determinant of technological diffusion – state-actors' alternate motivations may play a larger role in technology diffusion than previously thought.

References:

Abouk R, Pacula RL, Powell D. Association Between State Laws Facilitating Pharmacy Distribution of Naloxone and Risk of Fatal Overdose. JAMA Intern Med. 2019 Jun 1;179(6):805-811. doi: 10.1001/jamainternmed.2019.0272. PMID: 31058922; PMCID: PMC6503576.

Allen ST, O'Rourke A, Johnson JA, Cheatom C, Zhang Y, Delise B, Watkins K, Reich K, Reich R, Lockett C. Evaluating the impact of naloxone dispensation at public health vending machines in Clark County, Nevada. Ann Med. 2022 Dec;54(1):2692-2700. doi: 10.1080/07853890.2022.2121418. PMID: 36168975; PMCID: PMC9542801.

Centers for Disease Control and Prevention. (n.d.). *Multiple cause of death data on CDC Wonder*. Centers for Disease Control and Prevention. Retrieved December 9, 2022, from https://wonder.cdc.gov/mcd.html

Chimbar L, Moleta Y. Naloxone Effectiveness: A Systematic Review. J Addict Nurs. 2018 Jul/Sep;29(3):167-171. doi: 10.1097/JAN.0000000000000230. PMID: 30180002.

Cook Political. "Introducing the 2017 Cook Political Report Partisan Voter Index." *Cook Political Report*, https://www.cookpolitical.com/introducing-2017-cook-political-report-partisan-voter-index.

Davidson, P. J. "Fatal Heroin-Related Overdose in San Francisco, 1997-2000: A Case for Targeted Intervention." Journal of Urban Health: Bulletin of the New York Academy of Medicine, vol. 80, no. 2, 2003, pp. 261–273., https://doi.org/10.1093/jurban/jtg029.

Davis, C., Webb, D., & Burris, S. (2013). Changing Law from Barrier to Facilitator of Opioid Overdose Prevention. Journal of Law, Medicine & Ethics, 41(S1), 33-36. doi:10.1111/jlme.12035

Davis CS, Ruiz S, Glynn P, Picariello G, Walley AY. Expanded access to naloxone among firefighters, police officers, and emergency medical technicians in Massachusetts. Am J Public Health. 2014 Aug;104(8):e7-9. doi: 10.2105/AJPH.2014.302062. Epub 2014 Jun 12. PMID: 24922133; PMCID: PMC4103249.

Doleac, Jennifer L. and Mukherjee, Anita, The Effects of Naloxone Access Laws on Opioid Abuse, Mortality, and Crime (August 12, 2021). Available at SSRN: https://ssrn.com/abstract=3135264 or http://dx.doi.org/10.2139/ssrn.3135264

General, Office of the Surgeon. "U.S. Surgeon General's Advisory on Naloxone and Opioid Overdose." HHS.gov, 8 Apr. 2022, https://www.hhs.gov/surgeongeneral/reports-and-publications/addiction-and-substance-misuse/advisory-on-naloxone/index.html.

Griliches, Z. (1957). Hybrid Corn: An Exploration in the Economics of Technological Change. *Econometrica*, *25*(4), 501–522. https://doi.org/10.2307/1905380

Jones JD, Campbell A, Metz VE, Comer SD. No evidence of compensatory drug use risk behavior among heroin users after receiving take-home naloxone. Addict Behav. 2017 Aug;71:104-106. doi: 10.1016/j.addbeh.2017.03.008. Epub 2017 Mar 9. PMID: 28325710; PMCID: PMC5449215.

Kimball, Spence. “FDA Could Approve over-the-Counter Nasal Spray, Autoinjectors to Treat Opioid Overdoses.” CNBC, CNBC, 15 Nov. 2022, https://www.cnbc.com/2022/11/15/opioid-crisis-fda-may-approve-over-the-counter-naloxone-nasal-spray-autoinjectors.html.

Langham S, Wright A, Kenworthy J, Grieve R, Dunlop WCN. Cost-Effectiveness of Take-Home Naloxone for the Prevention of Overdose Fatalities among Heroin Users in the United Kingdom. Value Health. 2018 Apr;21(4):407-415. doi: 10.1016/j.jval.2017.07.014. Epub 2018 Feb 4. PMID: 29680097.

McClellan C, Lambdin BH, Ali MM, Mutter R, Davis CS, Wheeler E, Pemberton M, Kral AH. Opioid-overdose laws association with opioid use and overdose mortality. Addict Behav. 2018 Nov;86:90-95. doi: 10.1016/j.addbeh.2018.03.014. Epub 2018 Mar 19. PMID: 29610001.

Packham A, Syringe exchange programs and harm reduction: New evidence in the wake of the opioid epidemic, Journal of Public Economics, Volume 215, 2022, 104733, ISSN 0047-2727, https://doi.org/10.1016/j.jpubeco.2022.104733.

PDAPS - naloxone overdose prevention laws. (n.d.). Retrieved December 9, 2022, from https://pdaps.org/datasets/laws-regulating-administration-of-naloxone-1501695139

Rogers, E. M. (2005). *Diffusion of innovations*. Free Press.

Skinner, Jonathan, and Douglas Staiger. "Technology Adoption from Hybrid Corn to Beta-Blockers." In Hard-to-Measure Goods and Services: Essays in Honor of Zvi Griliches. Edited by Ernst R. Berndt, and Charles R. Hulten (eds). University of Chicago Press, 2007. Chicago Scholarship Online, 2013. https://doi.org/10.7208/chicago/9780226044507.003.0019.

Skinner, J., & Staiger, D. (2015). TECHNOLOGY DIFFUSION AND PRODUCTIVITY GROWTH IN HEALTH CARE. *The Review of Economics and Statistics*, *97*(5), 951–964. http://www.jstor.org/stable/43830289

Squire, P. (2017). A Squire Index Update. State Politics & Policy Quarterly, 17(4), 361–371. https://doi.org/10.1177/1532440017713314

Tracy, Melissa, et al. “Circumstances of Witnessed Drug Overdose in New York City: Implications for Intervention.” Drug and Alcohol Dependence, vol. 79, no. 2, 2005, pp. 181–190., https://doi.org/10.1016/j.drugalcdep.2005.01.010.

Wagner, K. D., et al. “Evaluation of an Overdose Preventionand Response Training Programme for Injection Drug Users int he Skid Row Area of Los Angeles, CA.” International Journal of Drug Policy, vol. 21,no. 3, 2010, pp. 186–193